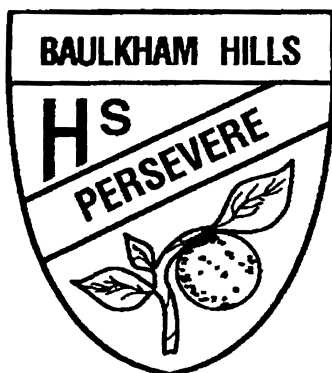




BAULKHAM HILLS HIGH SCHOOL

2013

YEAR 12 JUNE ASSESSMENT TASK

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 60minutes
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided
at the back of this paper
- All relevant mathematical reasoning
and/or calculations must be shown

Total marks – 34

There are 4 questions on pages 2-4.

Answer each question in the answer booklet. Each page must show your BOS#. Extra paper is available.

All relevant mathematical reasoning and/or calculations must be shown.

Marks

Question 1 Start on the relevant page in your answer booklet

1. a) Evaluate $\int_0^{\sqrt{3}} \frac{x}{\sqrt{x^2 + 1}} dx$ **2**

b) i) Find numbers a , b , and c such that **2**

$$\frac{2y + 3}{(y - 2)(y^2 + 3)} \equiv \frac{a}{y - 2} + \frac{by + c}{y^2 + 3}$$

ii) Hence find $\int \frac{2y + 3}{(y - 2)(y^2 + 3)} dy$ **2**

c) Evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \sin\theta + \cos\theta}$ **4**

Question 2 Start on the relevant page in your answer booklet

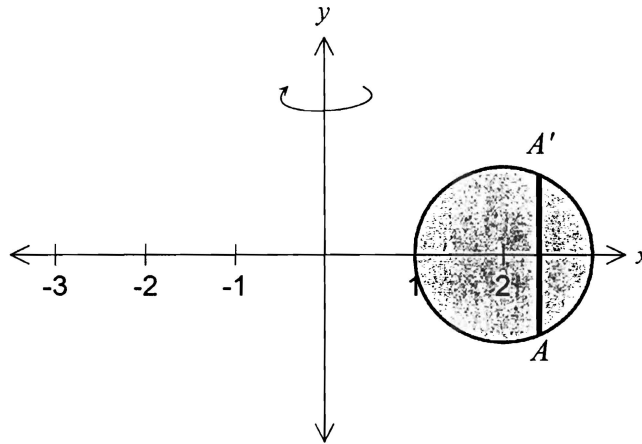
a) i) Prove $\int_0^a f(x)dx = \int_0^a f(a - x)dx$ **2**

ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \frac{1 - \tan x}{1 + \tan x} dx$ **3**

b) Find $\int \frac{x - 2}{\sqrt{8 - 2x - x^2}} dx$ **3**

Question 3 **Start on the relevant page in your answer booklet**

- a) The circular region $(x - 2)^2 + y^2 \leq 1$ is rotated about the y axis to form a torus.



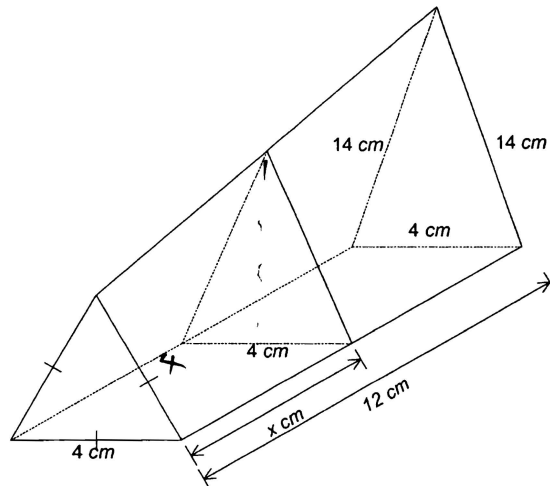
- i) Show that the volume ΔV obtained when a typical strip of height AA' and thickness Δx is rotated about the y axis is given by **2**

$$\Delta V = 4\pi x \sqrt{1 - (x - 2)^2} \Delta x$$

- ii) Hence find the total volume of the solid generated. **3**

- b) The front face of a solid is an equilateral triangle with sides 4 cm and the end face is an isosceles triangle with base 4 cm and equal sides 14 cm .

The solid is 12 cm long and cross sections parallel to the front face are isosceles triangles with base 4 cm .



- i) Show that the height, $h \text{ cm}$, of a triangular cross section $x \text{ cm}$, from the front face is given by $h = \frac{\sqrt{3}}{2}(x + 4)$ **3**

- ii) Hence find the volume of the solid. **2**

Question 4 Start on the relevant page in your answer booklet

Let $I_n = \int_0^1 \frac{x^n}{\sqrt{9-x^2}} dx$ where n is a positive integer.

i) Find the value of I_0 **1**

ii) Show that $nI_n = 9(n-1)I_{n-2} - 2\sqrt{2}$ for $n \geq 2$. **3**

iii) Hence evaluate $\int_0^1 \frac{x^4}{\sqrt{9-x^2}} dx$ **2**

END OF EXAMINATION

①

EXTENSION 2 JUNE 2013 SOLUTIONS

1 a)

$$\int_0^8 \frac{x dx}{\sqrt{x^2+1}}$$

$$= \frac{1}{2} \int_1^4 \frac{du}{\sqrt{u}} \quad \checkmark$$

$$= \frac{1}{2} \int_1^4 u^{-\frac{1}{2}} du$$

$$= \left[\sqrt{u} \right]_1^4$$

$$= \sqrt{4} - \sqrt{1}$$

$$= 1 \quad \checkmark$$

let $u = x^2 + 1$
 $du = 2x dx$
 when $x=3, u=4$
 $x=0, u=1$

2 marks correct answer
 1 mark significant progress

1 b i) let $\frac{2y+3}{(y-2)(y^2+3)} \equiv \frac{a}{y-2} + \frac{by+c}{y^2+3}$

$$2y+3 \equiv a(y^2+3) + (by+c)(y-2)$$

2 - correct answer
 1 - one correct or
 significant progress

sub $y=2$ $7=7a$
 $a=1$
 equating coefficients of y :
 $0 = a+b$
 $0 = 1+b$
 $b = -1$
 " of y^2 :
 $2 = -2b+c$
 $2 = 2+c$
 $c=0$
 $\therefore a=1, b=-1, c=0$

ii) $\int \frac{2y+3}{(y-2)(y^2+3)} dy = \int \frac{1}{y-2} - \frac{y}{y^2+3} dy$

$$= \ln|y-2| - \frac{1}{2} \ln(y^2+3) + C$$

2 correct answer
 1 - one correct integral

②

1 c. $\int_0^{\frac{\pi}{4}} \frac{d\theta}{1+\sin\theta+\cos\theta}$

$$= \int_0^1 \frac{2dt}{(1+t^2)\left(1+\frac{2t}{1+t^2}+\frac{1-t^2}{1+t^2}\right)} \quad \checkmark$$

$$= \int_0^1 \frac{2dt}{1+t^2+2t+1-t^2}$$

$$= \int_0^1 \frac{2dt}{2(t+1)}$$

$$= \int_0^1 \frac{dt}{t+1} \quad \checkmark$$

$$= \left[\ln(t+1) \right]_0^1$$

$$= \ln 2 - \ln 1$$

$$= \ln 2 \quad \checkmark$$

let $t = \tan \frac{\theta}{2}$
 $dt = \frac{1}{2} \sec^2 \frac{\theta}{2} d\theta$
 $2dt = (1+t^2) d\theta$
 $\therefore d\theta = \frac{2dt}{1+t^2} \quad \checkmark$
 when $\theta = \frac{\pi}{4}$ $t = \tan \frac{\pi}{4}$
 $t = 1$
 when $\theta = 0$, $t = \tan 0$
 $t = 0$

2 a i) $\int_a^b f(k-x) dx$

$$= \int_a^b f(u) (-du) \quad \checkmark$$

$$= \int_b^a f(u) du$$

$$= \int_a^b f(x) dx \quad \checkmark$$

let $u = a-x$
 $du = -dx$
 when $x=a$, $u=0$
 $x=b$, $u=a$

a ii) $\int_0^{\frac{\pi}{4}} \frac{1-\tan x}{1+\tan x} dx = \int_0^{\frac{\pi}{4}} \frac{1-\tan(\frac{\pi}{4}-x)}{1+\tan(\frac{\pi}{4}+x)} dx \quad \checkmark$

$$= \int_0^{\frac{\pi}{4}} \frac{1 - \frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \tan x}}{1 + \frac{\tan \frac{\pi}{4} + \tan x}{1 - \tan \frac{\pi}{4} \tan x}} dx$$

$$\begin{aligned}
 &= \int_0^{\frac{\pi}{4}} \frac{1 + \tan x - 1 - \frac{1}{2} + \tan x}{1 - \tan x + 1 + \tan x} dx \\
 &= \int_0^{\frac{\pi}{4}} \frac{2 \tan x}{2} dx \\
 &= \left[-\ln |\cos x| \right]_0^{\frac{\pi}{4}} \\
 &= -(\ln(\cos \frac{\pi}{4}) - \ln \cos 0) \\
 &= -\ln(\frac{1}{\sqrt{2}}) + \ln 1 \\
 &= \ln \sqrt{2} \\
 &= \frac{1}{2} \ln 2
 \end{aligned}$$

$$\begin{aligned} 2b) \quad \int \frac{x-2}{8-2x-x^2} dx &= \int \frac{x-2}{\sqrt{9-(x^2+2x+1)}} dx \\ &= \int \frac{x-2}{\sqrt{9-(x+1)^2}} dx \quad \checkmark \\ &= \frac{1}{2} \int \frac{-2(x+1)}{\sqrt{9-(x+1)^2}} dx - \int \frac{3}{\sqrt{9-(x+1)^2}} dx \\ &= \frac{1}{2} \int \frac{du}{\sqrt{u}} - 3 \sin^{-1} \frac{x+1}{3} + C \quad \checkmark \quad \begin{array}{l} \text{let } u = 9-(x+1)^2 \\ du = -2(x+1)dx \end{array} \\ &= -\sqrt{8-2x-x^2} - 3 \sin^{-1} \frac{x+1}{3} + C \end{aligned}$$

or method 2

$$\int \frac{x-2}{\sqrt{8-2x-x^2}} dx = \int \frac{x-2}{\sqrt{9-(x+1)^2}} dx$$

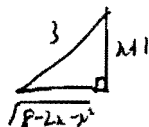
let $x+1 = 3 \sin \theta$
 $dx = 3 \cos \theta d\theta$
 $x-2 = 3 \sin \theta - 3$

$$= \frac{(3 \sin \theta - 3) 3 \cos \theta d\theta}{\sqrt{9-9 \sin^2 \theta}}$$

$$= \frac{9 (\sin \theta - 1) \cos \theta d\theta}{3 \sqrt{1-\sin^2 \theta}}$$

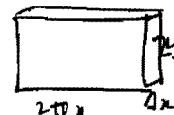
$$= 3 \int (\sin \theta - 1) d\theta$$

$$= 3 (-\cos \theta - \theta) + C$$

$$= -\sqrt{8-2x-x^2} - 3 \sin^{-1} \frac{x+1}{3} + C$$


④ 3. a) i)

$$\therefore M': \quad y^2 = \frac{1 - (x-2)^2}{2y} \quad (y > 0) \quad \checkmark$$



$$\therefore \Delta V = 2\pi h \cdot 2\sqrt{1-(h-2)^2} \Delta h$$

$$= 4\pi h \sqrt{1-(h-2)^2} \Delta h$$

ii) $V = \lim_{\Delta x \rightarrow 0} \sum_{k=1}^3 4\pi x \sqrt{1-(x-z)^2} \Delta x$ ** must start this line*

$= 4\pi \int_0^3 x \sqrt{1-(x-z)^2} dx$ let $u = x-2$ $x = u+2$
 $du = dx$

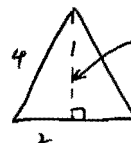
$= 4\pi \int_{-1}^1 (u+2) \sqrt{1-u^2} du$ ✓ when $x=3$, $u=1$
 $x=1$ $u=-1$

$= 4\pi \left(\int_{-1}^1 \underbrace{u \sqrt{1-u^2}}_{\substack{\uparrow \\ \text{odd function}}} + 2 \int_{-1}^1 \underbrace{\sqrt{1-u^2}}_{\substack{\uparrow \\ \text{even function}}} du \right)$

$= 4\pi \left[0 + 2 \int_{-1}^1 \sqrt{1-u^2} du \right]$

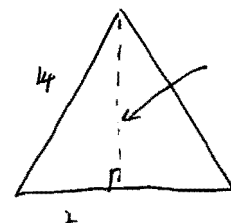
$= 4\pi \times 2 \times \frac{\pi \times 1^2}{2}$

Volume = $4\pi^2 \text{ units}^3$ ✓



$$h_F = \sqrt{16 - 4}$$

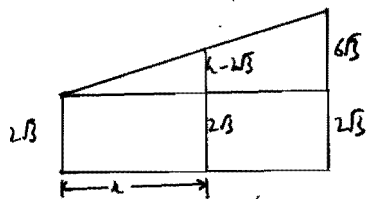
$$= 2\sqrt{3}$$



$$h_2 = \sqrt{14^2 - 2^2}$$
$$= 8\sqrt{3}$$

3 b)

method 1 Similarity



$$\frac{x-2\sqrt{3}}{6\sqrt{3}} = \frac{x}{12} \quad \checkmark \text{ as triangles are similar}$$

$$x-2\sqrt{3} = \frac{x\sqrt{3}}{2}$$

$$x = \frac{\sqrt{3}}{2} (x+4)$$

$$x = \frac{\sqrt{3}}{2} (x+4) \quad \checkmark$$

method 2 Linear functions

$$\text{let } h = ax + b$$

$$\text{when } x=0, h=2\sqrt{3}$$

$$2\sqrt{3} = 0 + b$$

$$\therefore b = 2\sqrt{3}$$

$$\text{when } x=12, h=8\sqrt{3}$$

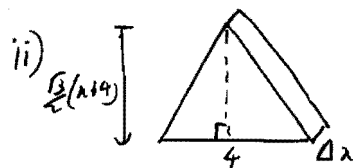
$$8\sqrt{3} = 12a + 2\sqrt{3}$$

$$6\sqrt{3} = 12a$$

$$a = \frac{\sqrt{3}}{2}$$

$$\therefore h = \frac{\sqrt{3}}{2} x + 2\sqrt{3}$$

$$= \frac{\sqrt{3}}{2} (x+4) \quad \checkmark$$



$$\text{Volume of slice } \Delta V = \frac{1}{2} \times 4 \times \frac{\sqrt{3}}{2} (x+4) \Delta x$$

$$\Delta V = \sqrt{3} (x+4) \Delta x$$

$$\text{Volume} = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^n \sqrt{3} (x+4) \Delta x$$

$$= \sqrt{3} \int_0^{12} (x+4) dx$$

$$= \sqrt{3} \left[\frac{x^2}{2} + 4x \right]_0^{12}$$

$$= \sqrt{3} (72 + 48 - 0)$$

$$\therefore \text{Volume} = 120\sqrt{3} \text{ cm}^3 \quad \checkmark$$

$$4 \text{ i)} \quad I_0 = \int_0^1 \frac{x^2}{\sqrt{9-x^2}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{9-x^2}} dx$$

$$= \left[\sin^{-1} \frac{x}{3} \right]_0^1$$

$$I_0 = \sin^{-1} \frac{1}{3} \quad \checkmark$$

$$4 \text{ ii)} \quad I_n = \int_0^1 \frac{x^n}{\sqrt{9-x^2}} dx$$

$$= \int_0^1 \frac{x^{n-1}}{\sqrt{9-x^2}} dx$$

$$u = x^{n-1}$$

$$u' = \frac{x}{\sqrt{9-x^2}}$$

$$I_n = \left[-x^{n-1} \sqrt{9-x^2} \right]_0^1 + (n-1) \int_0^1 x^{n-2} \sqrt{9-x^2} dx$$

$$= (-\sqrt{8} - 0) + (n-1) \int_0^1 \frac{x^{n-2} (9-x^2)}{\sqrt{9-x^2}} dx$$

$$I_n = -2\sqrt{2} + (n-1) \int_0^1 \frac{9x^{n-2} - x^n}{\sqrt{9-x^2}} dx$$

$$I_n = -2\sqrt{2} + (n-1) \int_0^1 \frac{9x^{n-2}}{\sqrt{9-x^2}} dx - (n-1) \int_0^1 \frac{x^n}{\sqrt{9-x^2}} dx$$

$$I_n = -2\sqrt{2} + 9(n-1) \int_0^1 \frac{x^{n-2}}{\sqrt{9-x^2}} dx - (n-1) I_n$$

$$I_n (1 + n - 1) = 9(n-1) I_{n-2} - 2\sqrt{2}$$

$$n I_n = 9(n-1) I_{n-2} - 2\sqrt{2}$$

$$4 \text{ iii)} \quad I_4 = \int_0^1 \frac{x^4}{\sqrt{9-x^2}} dx$$

$$4 I_4 = 9 \times 3 I_2 - 2\sqrt{2}$$

$$4 I_4 = \frac{27}{2} I_2 - 2\sqrt{2}$$

$$= \frac{27}{2} (9 I_0 - 2\sqrt{2}) - 2\sqrt{2}$$

$$4 I_4 = \frac{27}{2} \left(9 \sin^{-1} \left(\frac{1}{3} \right) - 2\sqrt{2} \right) - 2\sqrt{2}$$

$$I_4 = \frac{27}{8} \left(9 \sin^{-1} \left(\frac{1}{3} \right) - 2\sqrt{2} \right) - \frac{\sqrt{2}}{2}$$

$$I_4 = \frac{243}{8} \sin^{-1} \left(\frac{1}{3} \right) - \frac{29\sqrt{2}}{4} \quad \checkmark$$